

Lecture 03

Coasean Bargaining

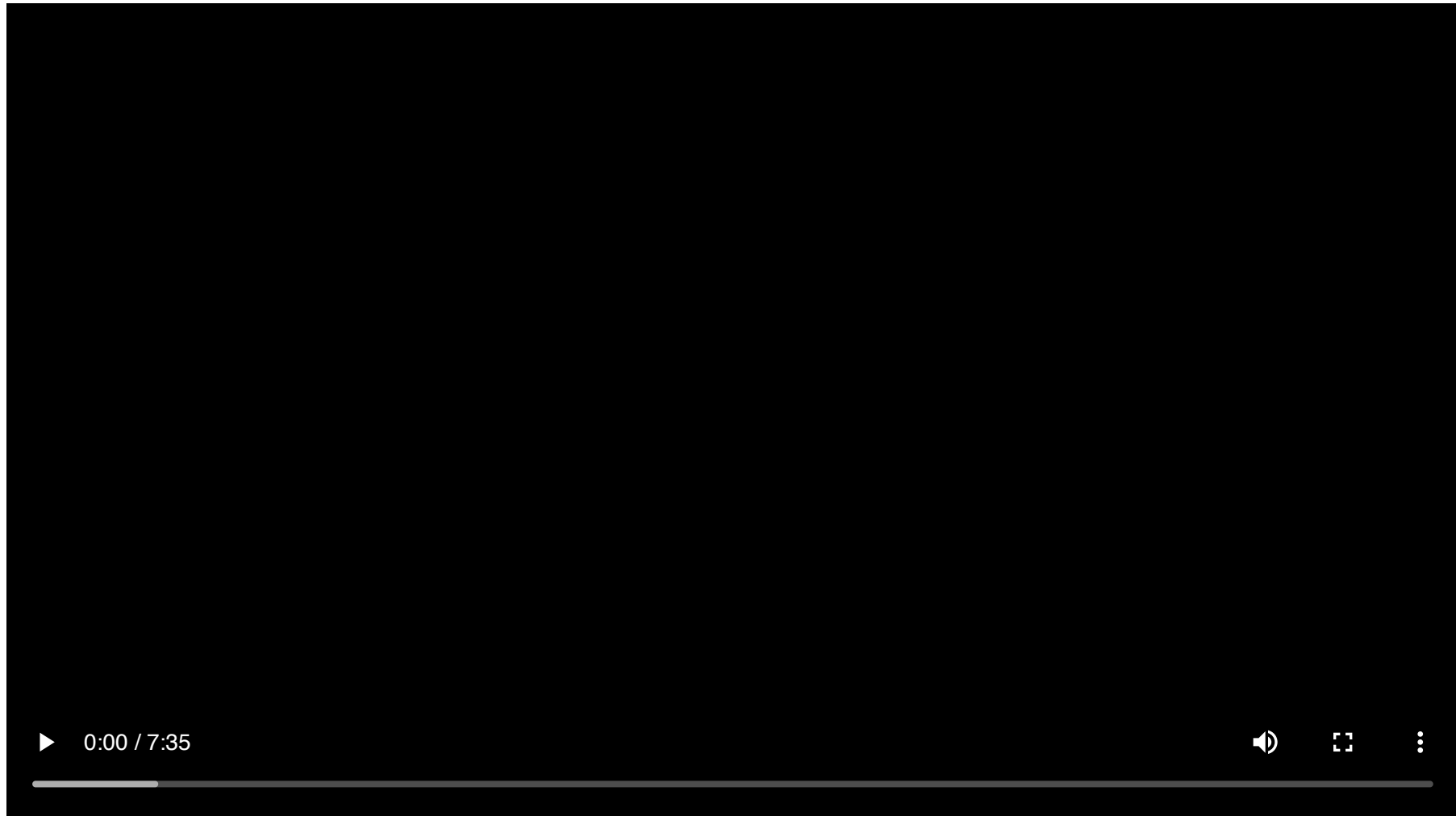
Ivan Rudik

AEM 4510

Roadmap

1. Can we achieve the efficient outcome **without** government intervention?
2. What does the Coase theorem say?
3. What are the limits to Coasean bargaining?
4. Can data centers and their neighbors bargain their way out of the AI boom's externalities?

This lecture's running example: the AI boom next door



Data centers are a textbook externality problem

Data centers bring jobs and tax revenue, and their neighbors bear real costs:

- Cooling fans and diesel backup generators hum around the clock
- Generator testing burns diesel next to homes
- Water use competes with local supplies
- Electricity demand pushes up everyone's bills

The parties are easy to name: one operator, identifiable neighbors

Today's question: when can the operator and the neighbors bargain to the efficient outcome, and when do transactions costs get in the way?

Coase Theorem

Pigou vs Coase

We have argued that efficient allocations are not achieved in the presence of externalities

Why?

There are no markets through which the source of the externality must pay/be compensated for its effect on society

i.e. they're not priced

This means there's a role for government to create this market or price the externality

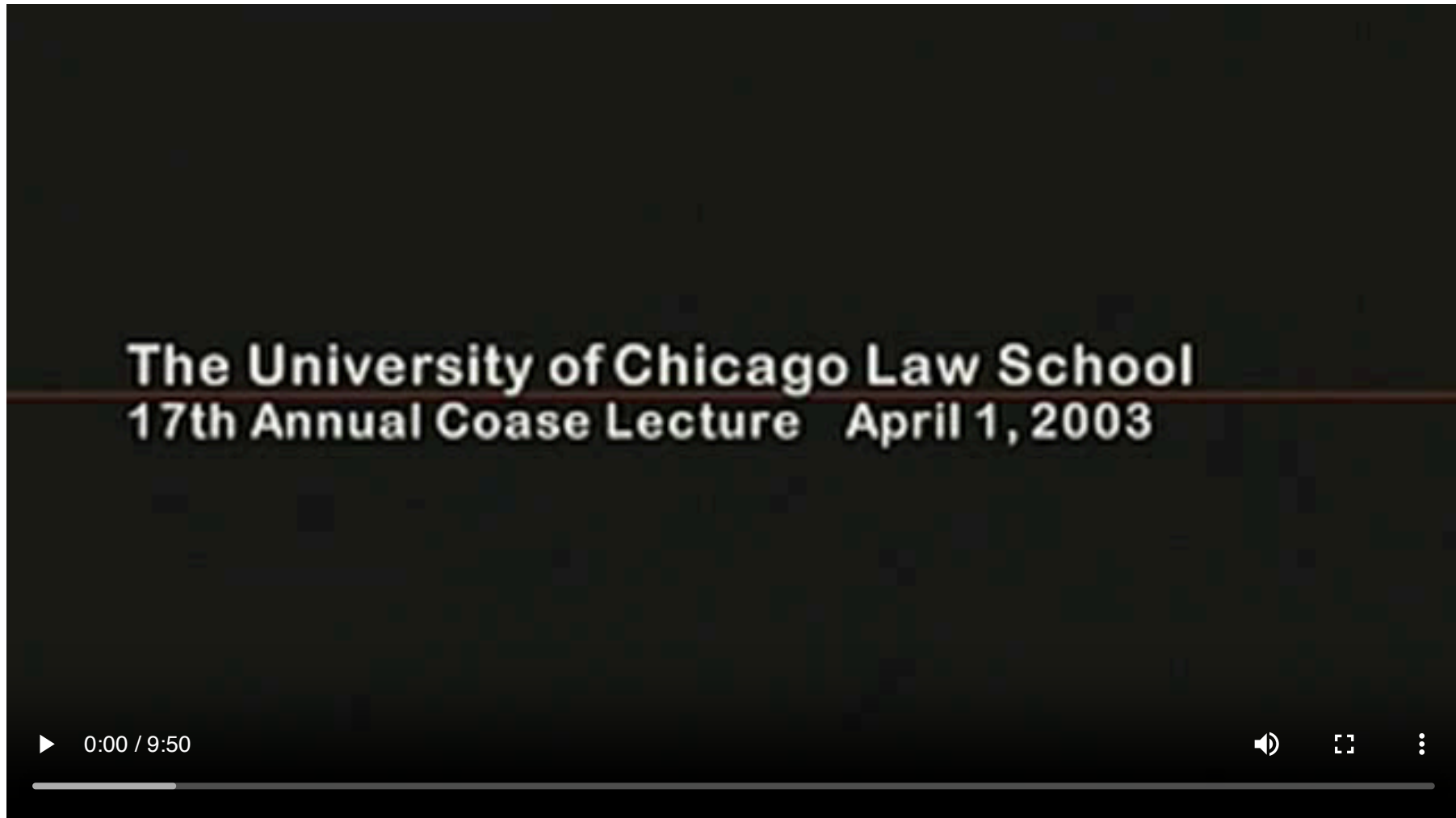
Ronald Coase (1910-2013)



In a famous paper ("The Problem of Social Cost"), 1991 Nobel prize winner Ronald Coase made people rethink this

Do we actually **NEED** government intervention?

Ronald Coase (1910-2013)



The Coase Theorem

If there are:

1. No wealth effects on demand
2. No transactions costs
3. Well-defined and enforceable property rights

then:

the socially efficient or optimal economic activity will occur regardless of who holds the property rights

The right to pollute (a resource) will end up in the hands who value it most through negotiation

Property rights



Property rights are only as good as prevailing norms or enforcement

Coasean arguments

Coase versus Pigou: externalities are reciprocal in nature

A power plant produces emissions that nearby residents breathe and those people incur the external costs

By breathing the air, the nearby residents help create the externality (i.e. if they weren't there, there would be no external cost from emitting pollution)

Coasean arguments

What is more valuable to society?

The externality-producing good generated by the power plant or letting people live nearby?

Argument resonates better in the context of the legal cases being considered by Coase (e.g. the doctor and the confectioner). In the context of the power plant the victims aren't "producing" anything

The Doctor and the Confectioner

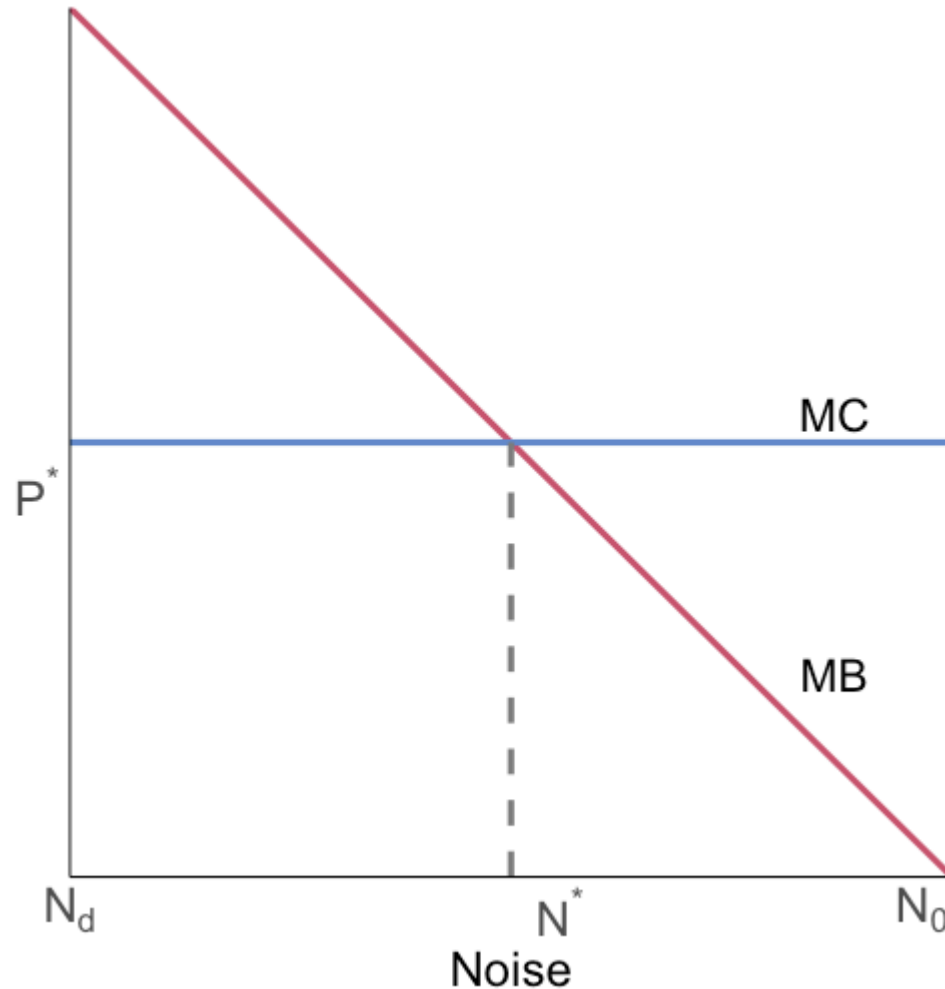
More noise = more candy and less medical services

Less noise = less candy and more medical services

Which is better from a social point of view depends upon the relative values of candy and medical services

Is the net benefit to society better at no noise, 0, or the level of noise that maximizes confectioner profit, N_0

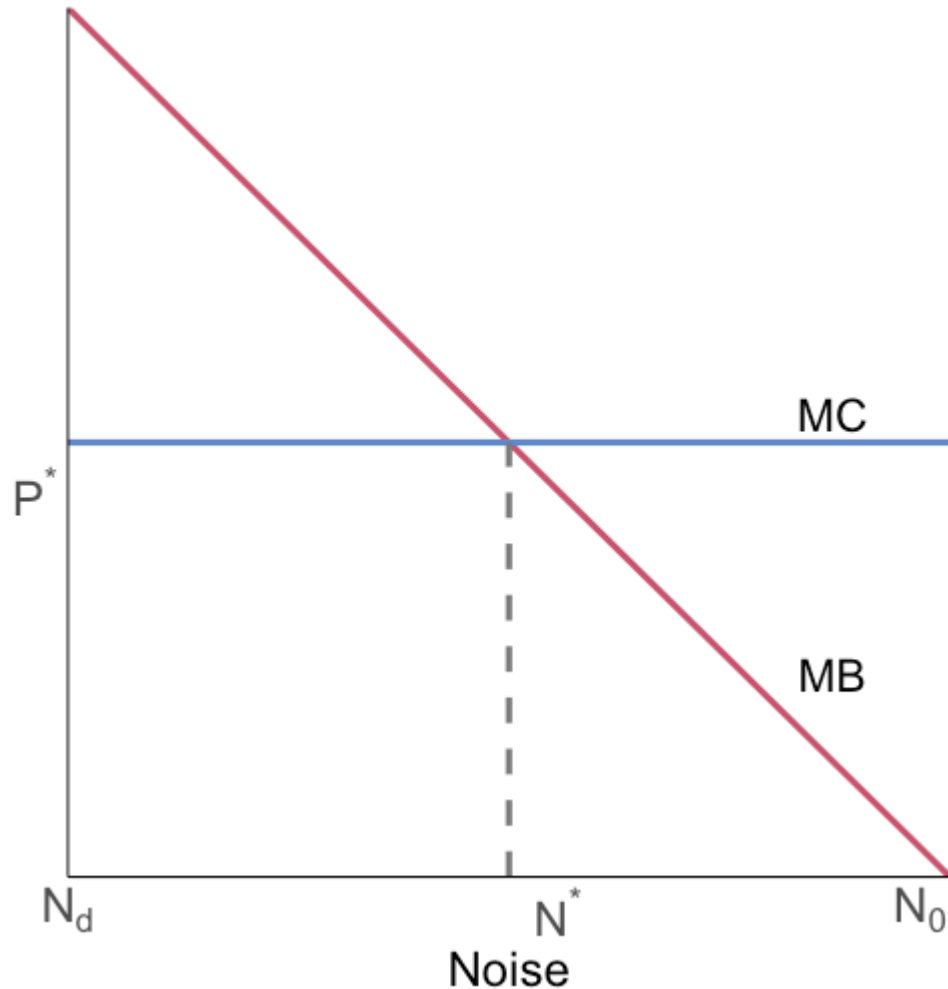
The Doctor and the Confectioner



MC is the marginal cost imposed on the doctor by noise

MB is the marginal benefit to the confectioner (marginal profits) from the production process that creates noise

Coase: Point 1



It is important to establish that someone has the property rights

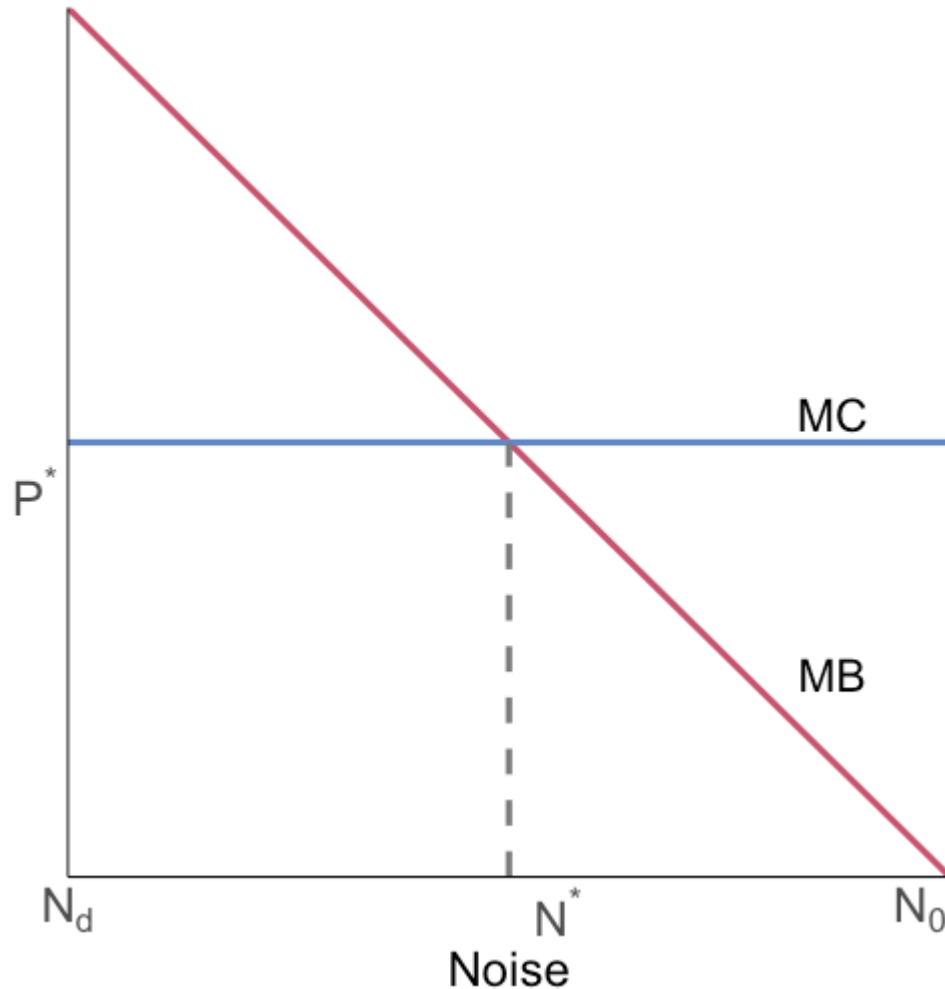
Otherwise, trade will not happen

Give property rights to the confectioner

Initial outcome will be $N=N_0$

What happens next?

Coase: Point 1

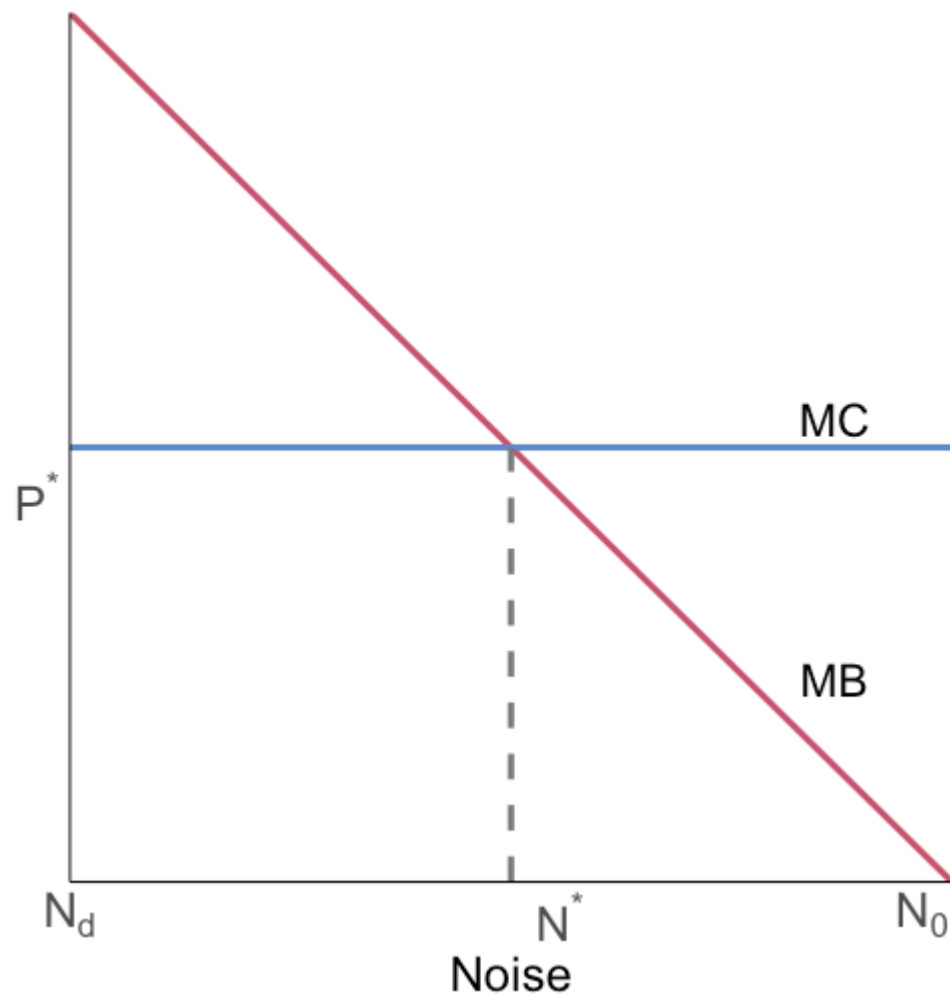


the doctor can pay the confectioner to stay quiet (stop producing) for part of the day

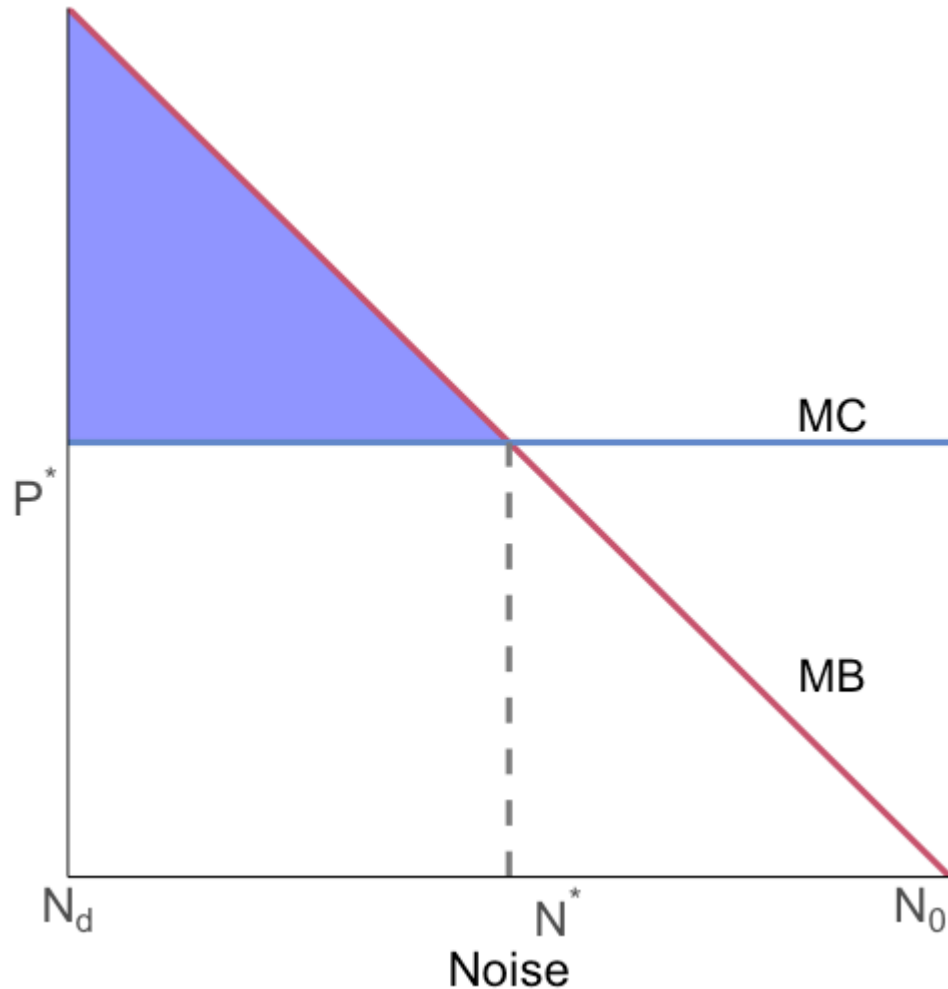
Why?

Because MC to the doctor is higher than the MB to the confectioner for the units of noise after N^*

Coase: Point 1



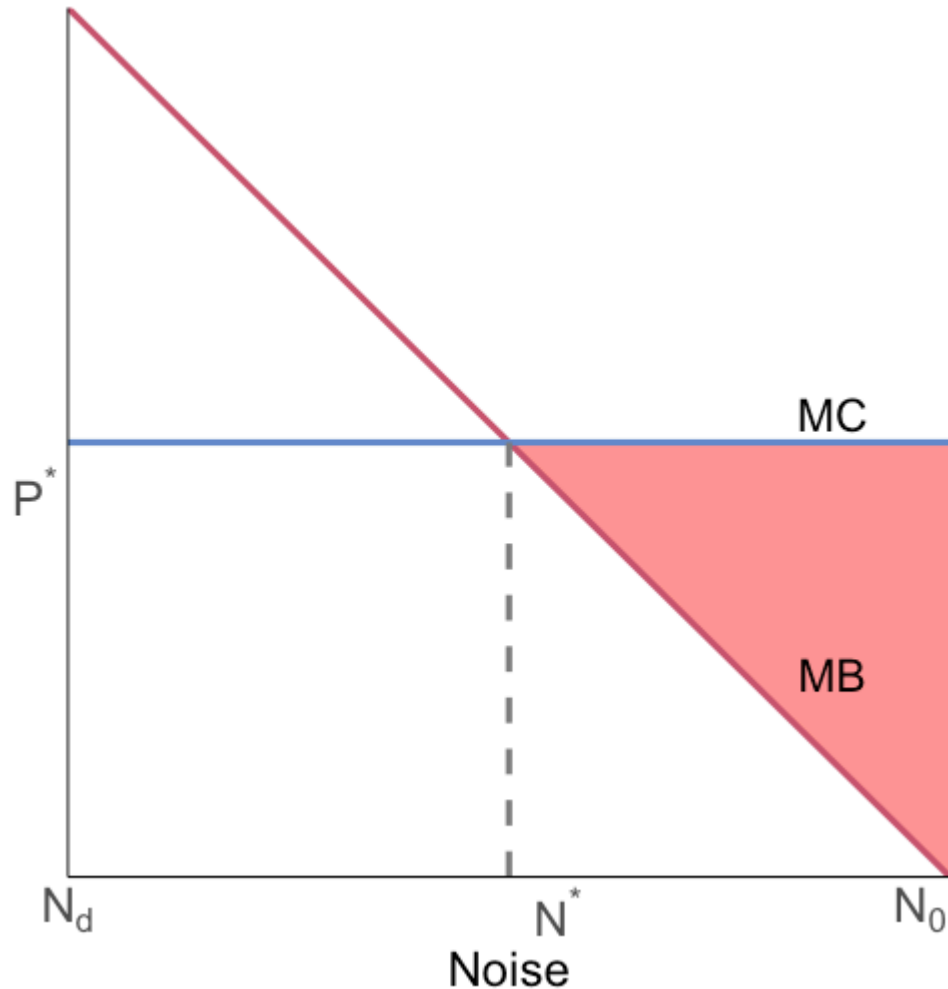
Coase: Point 1



The doctor is willing to pay more (MC) than the confectioner is willing to accept (MB) until noise is reduced to N^*

This is where total benefit is maximized (blue area)

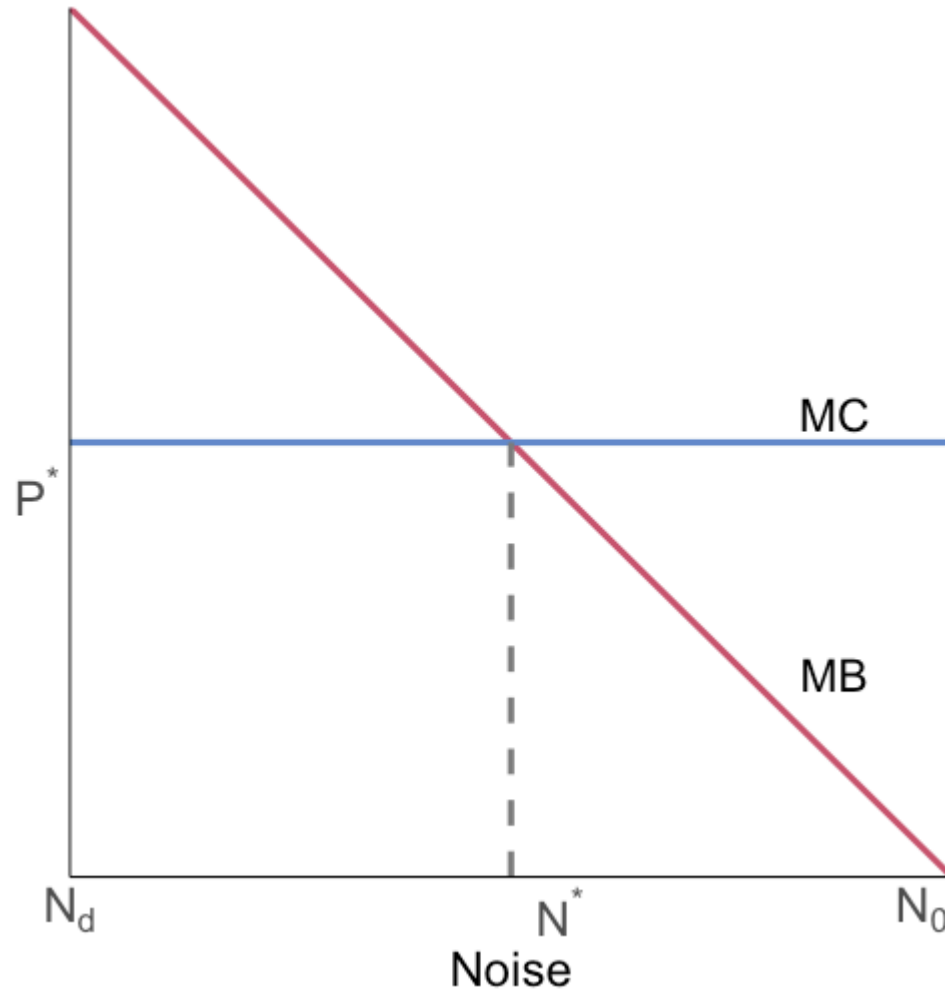
Coase: Point 1



The doctor and confectioner can split the **bargaining surplus**, the red area

This is just the avoided deadweight loss from the noise externality

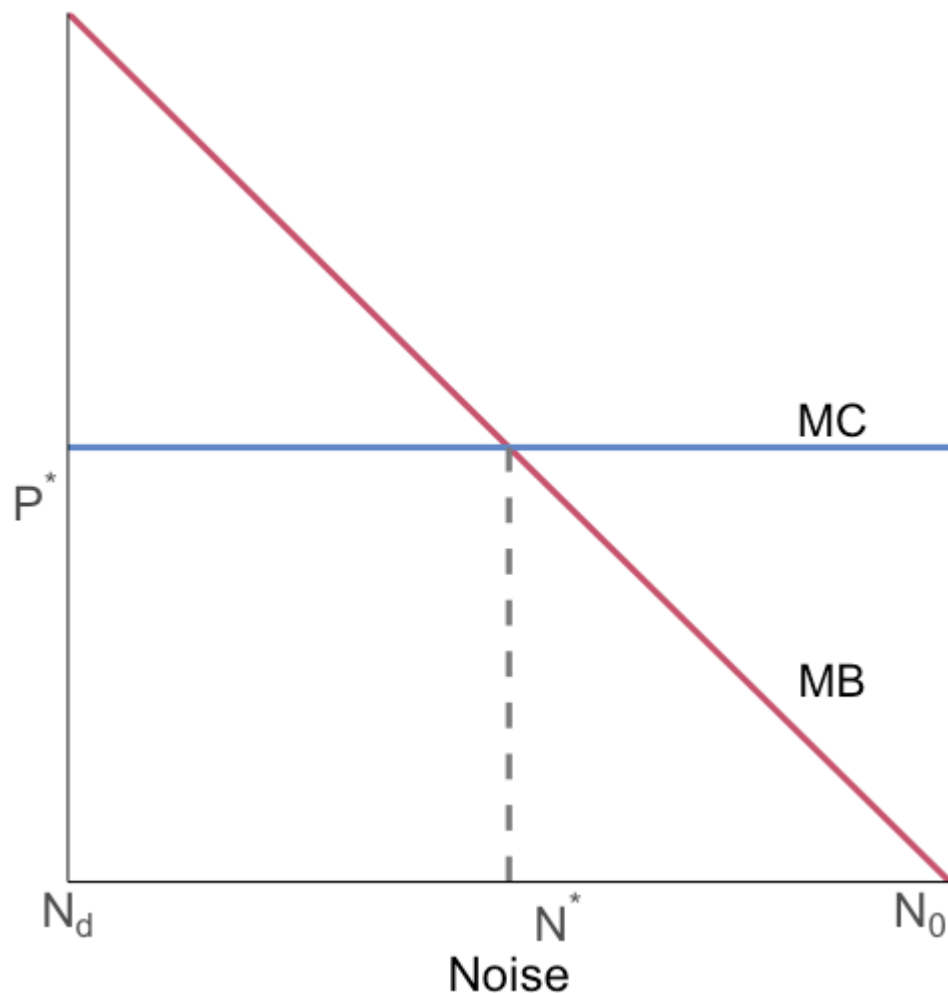
Coase: Point 1



Instead of assigning property rights to the confectioner we could have assigned them to the doctor

In this case what happens?

Coase: Point 1

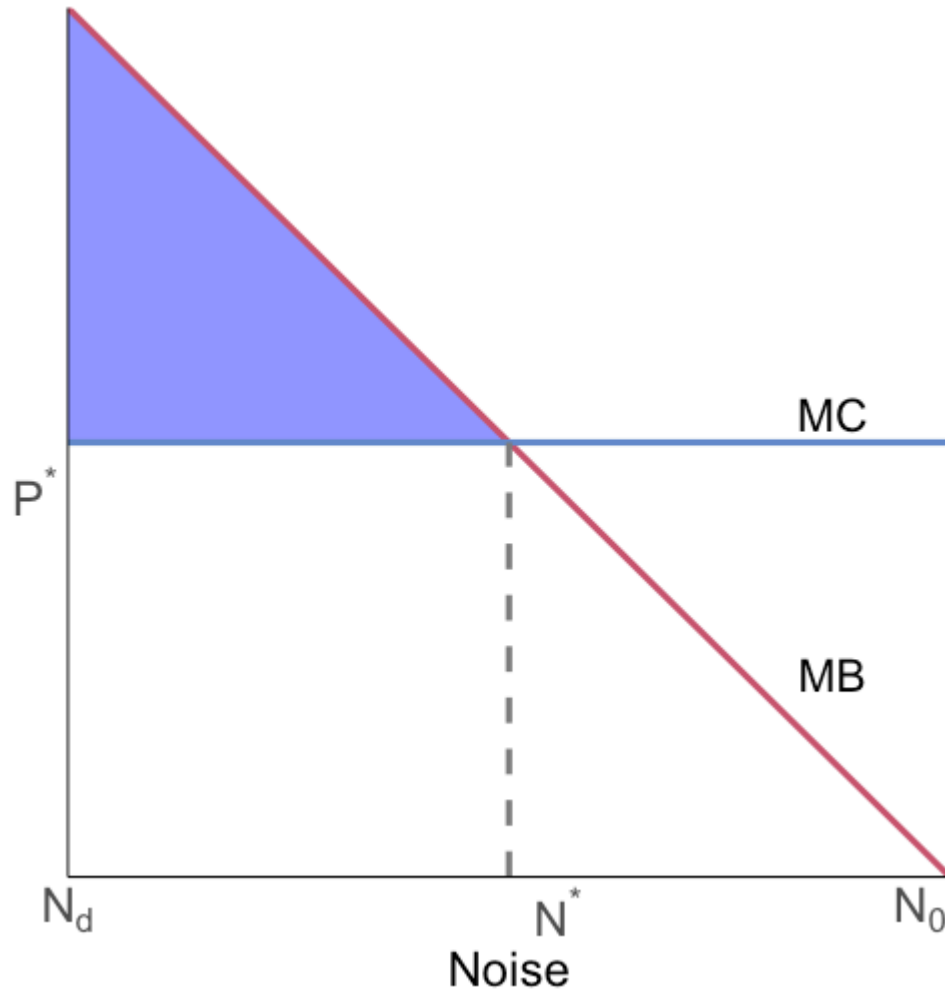


First, we start at N_d now since the doctor does not like noise

Confectioner pays the doctor to be allowed to make noise

The confectioner is willing to pay (MB) more than the doctor is willing to accept (MC) until we reach N^*

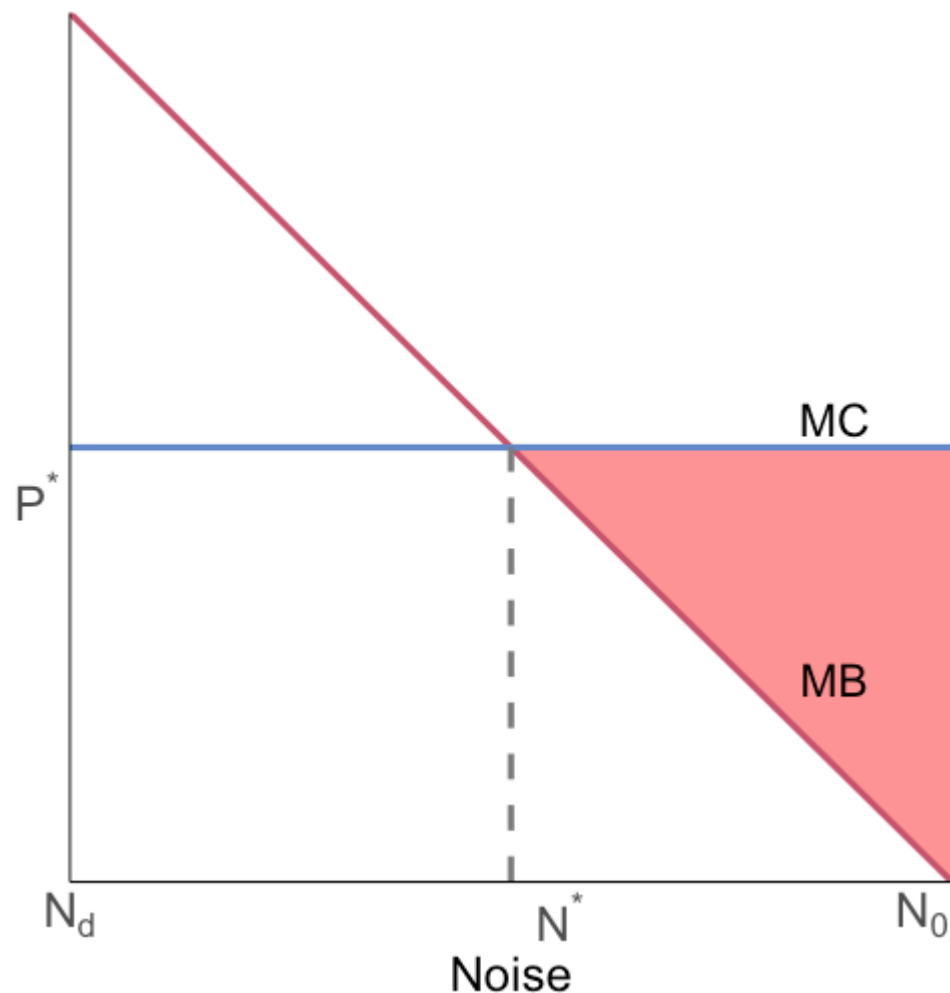
Coase: Point 1



We now maximize surplus (**blue**) and gain bargaining surplus (**blue**) that is split between the doctor and confectioner

It didn't matter who had the property rights, we managed to get to N^*

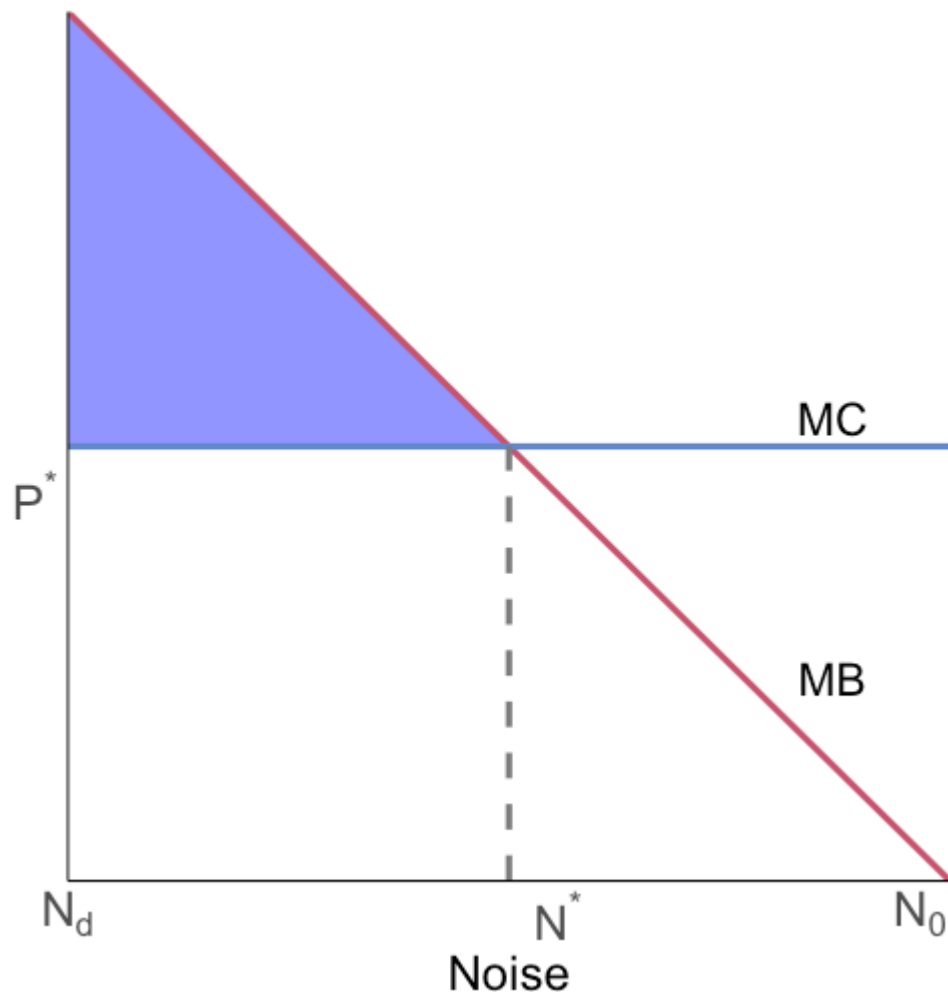
Coase: Point 2



The initial assignment of property rights does matter for the distribution of surplus

If we give the confectioner property rights, they get paid by the doctor some quantity up to the total size of the red area (bargaining surplus)

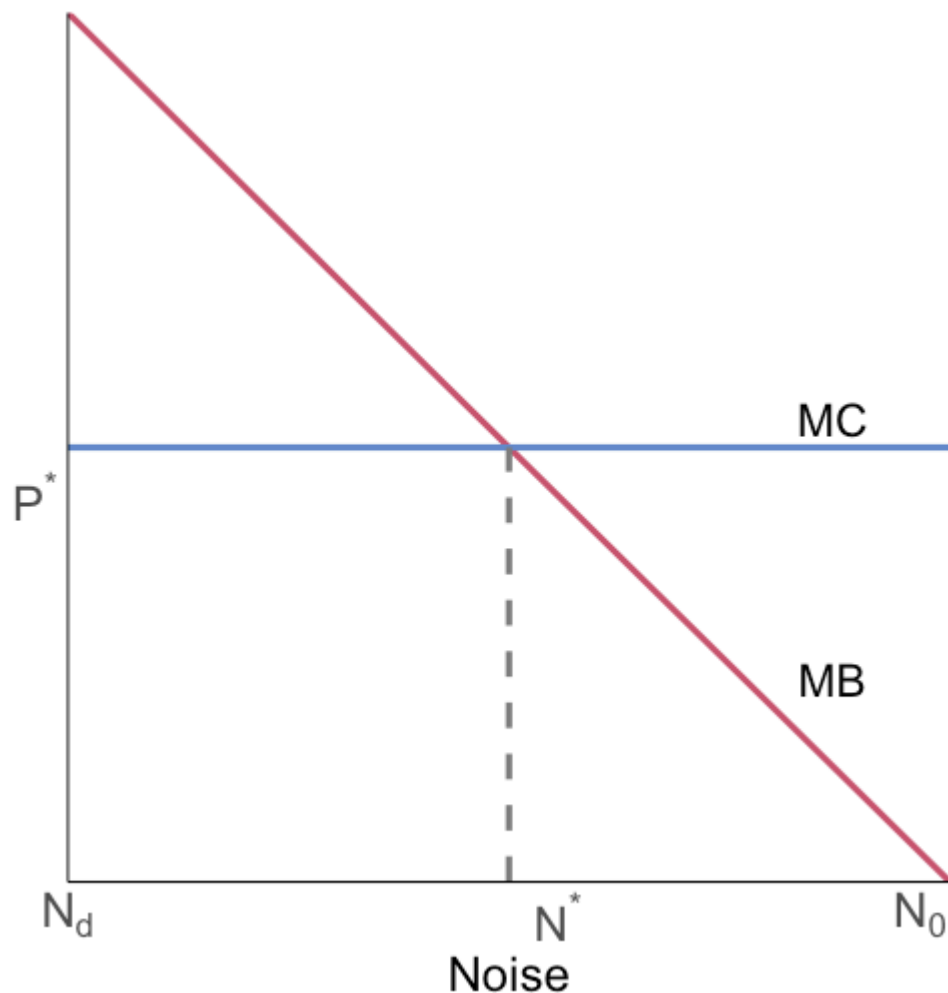
Coase: Point 2



The initial assignment of property rights does matter for the distribution of surplus

If we give the doctor property rights, they get paid by the confectioner some quantity up to the total size of the blue area (bargaining surplus)

Coase: Point 2



This means that property rights are valuable!

If you have property rights, others have to incentivize you in order to deviate from your privately optimal choice

You will only change the level of noise if your welfare/surplus improves

Coase: Point 3

What if the choice is discrete: noise or silence?

Suppose the surplus to the two people under noise and silence is given by:

	Confectioner	Doctor
Noise	500	0
Silence	0	250

What happens?

Total surplus is maximized with noise ($500 > 250$)...

Coase: Point 3

	Confectioner	Doctor
Noise	500	0
Silence	0	250

If the confectioner has the property rights, we are already at the efficient outcome

If the doctor has property rights, the confectioner can pay the doctor > 250 but < 500 and both are better off, a Pareto improvement!

Coase Caveats

Coasean bargaining does not always work like you think

There are two key pieces we need to have satisfied:

1. No Transactions costs
2. No wealth/income effects

Coase Caveats: Transactions costs

Suppose the doctor owns the property right of zero noise

Noise N imposes cost $C(N)$ on the doctor, benefits $B(N)$ to the confectioner

Status quo is no contract: no additional noise and no transfer payment

This is a one-shot game: the confectioner gets one take-it-or-leave-it offer where the doctor accepts some noise N in exchange for payment θ

Coase Caveats: Transactions costs

Timing:

1. Confectioner pays transaction/setup cost tr to propose one contract (N, θ)
2. Doctor accepts or rejects once
3. If rejected, they stay at the status quo (no additional noise, no transfer)

There is no renegotiation after the accept/reject decision

Coase Caveats: Transactions costs

When does the doctor accept the contract?

The doctor is weakly better off accepting the contract if the transfer payment θ (the benefit) is at least the cost of noise $C(N)$

In our graphical example, $C(N) = MC \times N$ since MC is constant

What contract does the confectioner offer in equilibrium?

i.e. what contract proposal maximizes the confectioner's profit?

Coase Caveats: Transactions costs

The confectioner will choose to offer $\theta = MC \times N$

Why?

In a one-shot take-it-or-leave-it offer, the proposer sets the responder at reservation utility

So $\theta = C(N) = MC \times N$ is the least amount required for the doctor to accept

If bargaining were unlimited, for a given N the feasible theta range is $\theta \in [MC \times N, B(N)]$

The realized payment in that interval depends on bargaining power and

Coase Caveats: Transactions costs

This means we can write the confectioner's total profit as:

$$\pi(N) = B(N) - MC \times N - tr$$

Where $B(N)$ is the benefits to the confectioner of noise (area under MB)

It's optimal choice of N (and θ) is given by the first-order condition:

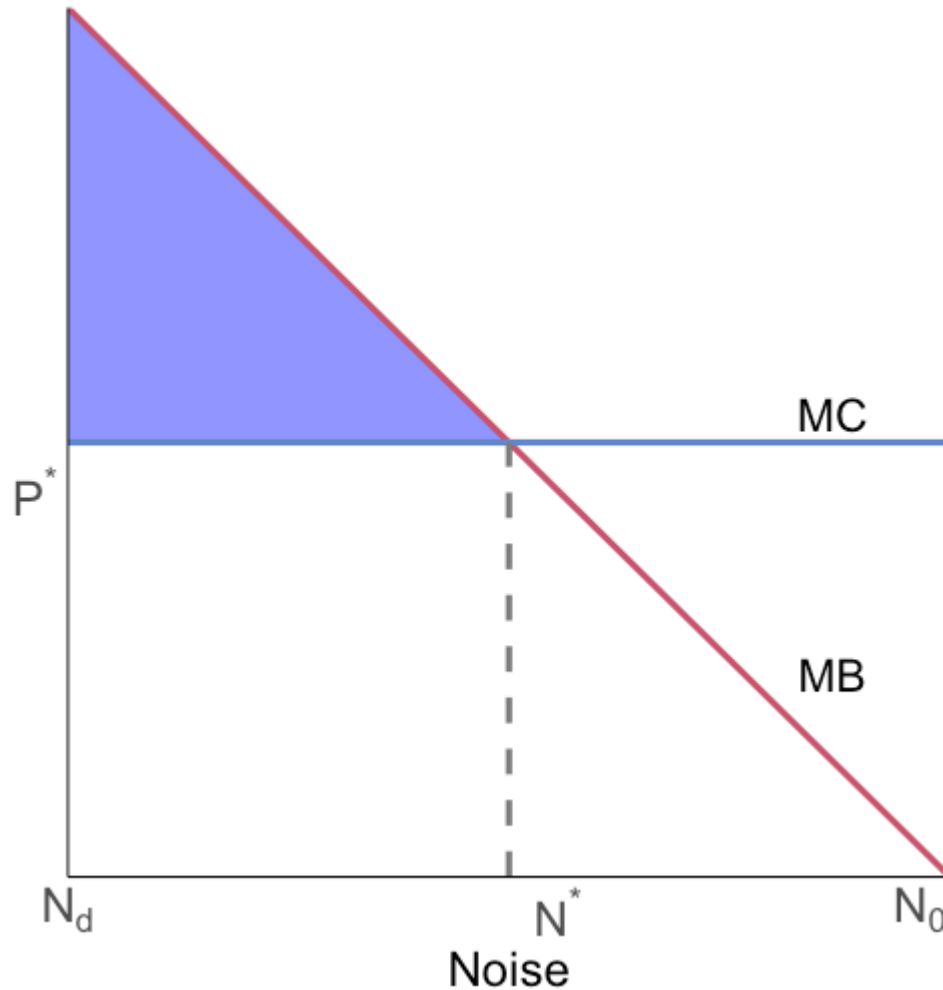
$$B'(N) = MC \leftrightarrow MB(N) = MC$$

which matches our condition for efficiency in noise

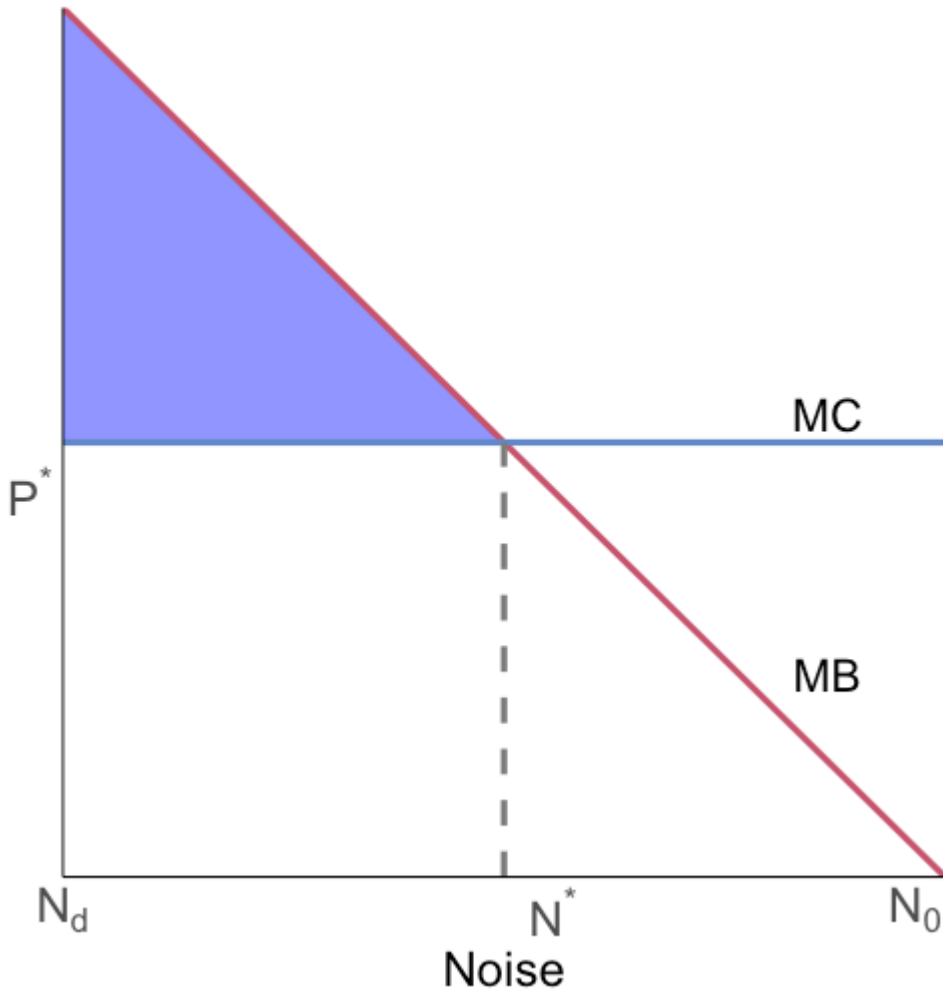
Did transactions costs actually cause any problems?

Coase Caveats: Transactions costs

Yes! Why?



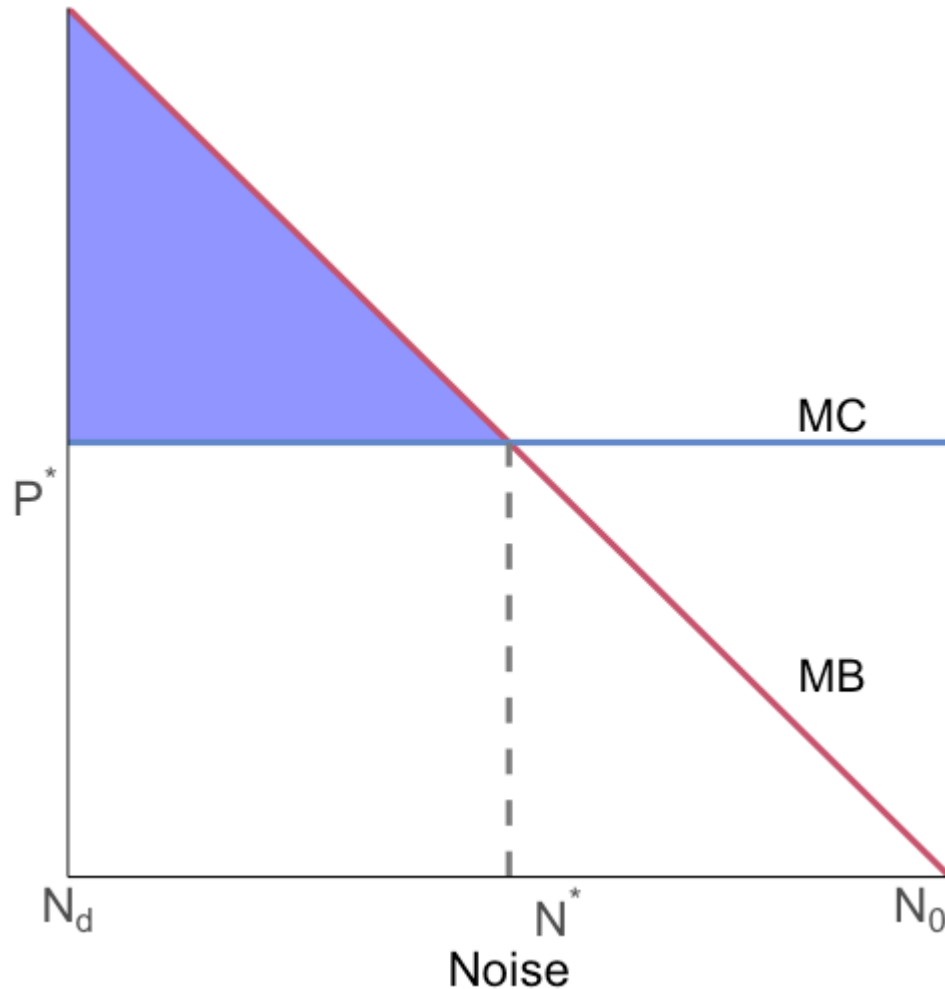
Coase Caveats: Transactions costs



Yes! Why?

To have a mutually beneficial contract we still need the total gain in surplus (**blue**) to be greater than tr

Coase Caveats: Transactions costs



Yes! Why?

To have a mutually beneficial contract we still need the total gain in surplus (**blue**) to be greater than tr

Otherwise the total cost of the bargaining is greater than the total benefit from bargaining → bargaining makes us worse off

Coase Caveats: Transactions costs

If the confectioner has the rights to noise, we just flip the script

The doctor makes a one-shot take-it-or-leave-it contract offer (N, θ) where the confectioner reduces noise from N_0 in exchange for a transfer payment

The confectioner accepts or rejects once; if rejected, they stay at N_0 with no transfer

When does the confectioner accept the contract?

The confectioner accepts if $\theta \geq B(N_0) - B(N)$ (payment $>$ loss of benefits)

Under one-shot take-it-or-leave-it bargaining, the doctor offers the minimum required: $\theta = B(N_0) - B(N)$

Coase Caveats: Transactions costs

The doctor's problem is then to maximize the benefits of the transaction:

$$\max_N \underbrace{MC \times (N_0 - N)}_{\text{noise cost reduction}} - \theta - tr$$

Which we can write as:

$$\max_N MC \times (N_0 - N) - \underbrace{(B(N_0) - B(N))}_{\theta} - tr$$

Coase Caveats: Transactions costs

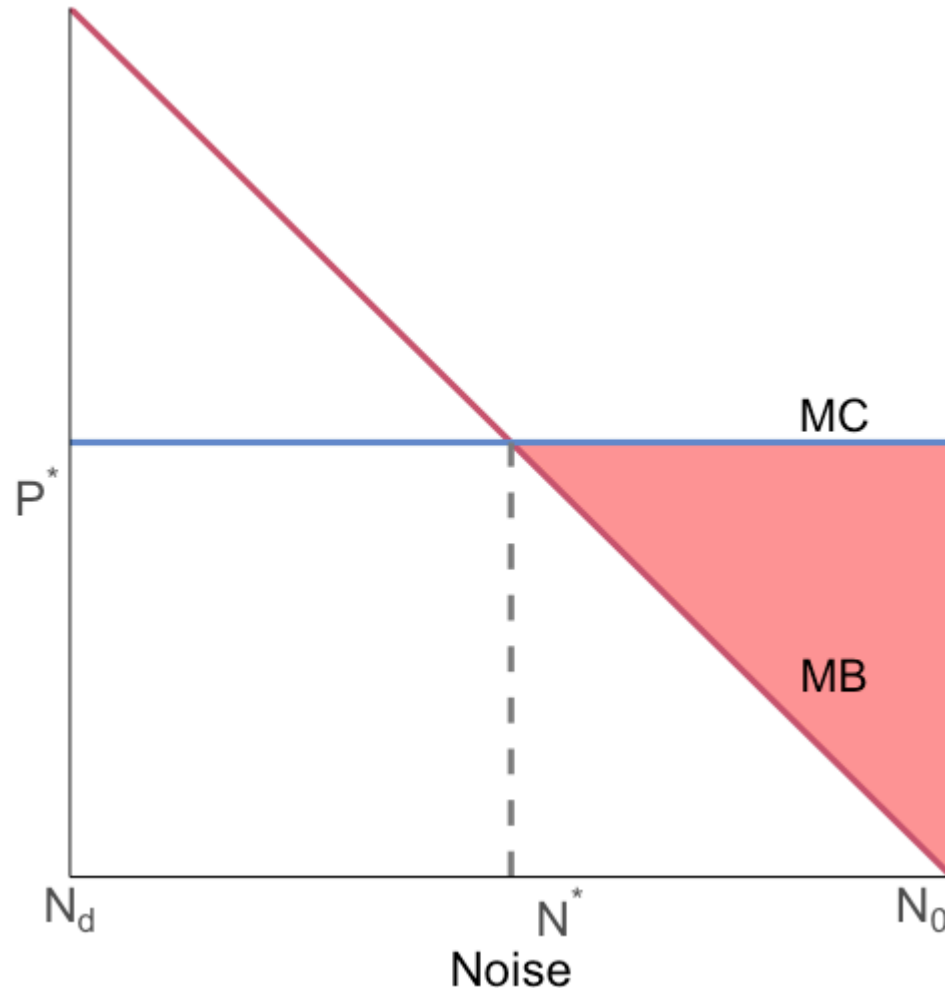
$$\max_N MC \times (N_0 - N) - \underbrace{(B(N_0) - B(N))}_{\theta} - tr$$

First term is the avoided noise costs, second term is the payment, third term is the transaction cost

The doctor's problem gives us the first-order condition:

$$MC = B'(N) \equiv MC = MB$$

Coase Caveats: Transactions costs



We again reach the social optimum!

To have a mutually beneficial contract we still need the total gain in surplus (**red**) to be greater than tr

Otherwise the total cost of the bargaining is greater than the total benefit from bargaining → bargaining makes us worse off

Coase Caveats: Transactions costs

Main takeaway: transactions costs can prevent Coasean bargaining from achieving the efficient allocation

Why?

In cases where the gains from bargaining are small, transactions costs may exceed the benefits and prohibit bargaining from occurring

Coase Caveats: Wealth effects

Changing the initial endowment can change our outcome if wealth changes the doctor's preferences for noise

The property right is valuable: the owner is richer even before trading

Quiet makes the doctor more productive: owning the property right raised his wealth before negotiating

With diminishing marginal utility of wealth, extra dollar matters less:

- If the doctor owns quiet, he requires a larger payment to accept noise
- If he does not own quiet, he is not willing to pay as much to reduce the noise

Coase Caveats: Wealth effects

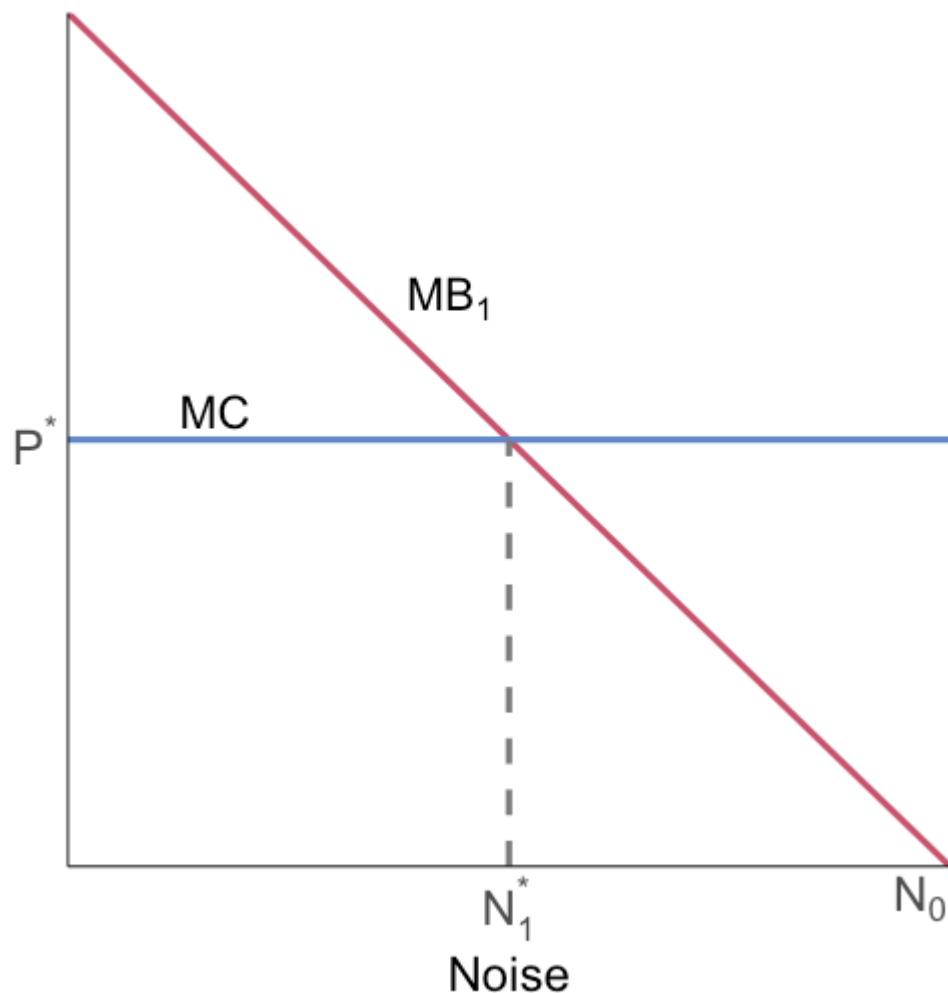
Suppose the confectioner gains \$90 from noise

When the doctor owns the right to quiet he is wealthier and values the noise at \$100, but when the doctor does not own the right to quiet he is poorer and values the noise at \$80

- Doctor owns quiet (wealthier): WTA to accept noise = \$100 → no trade → quiet
- Confectioner owns noise (doctor poorer): WTP for quiet = \$80 → no trade → noise

Different initial rights → different final allocation

Coase Caveats: Income effects



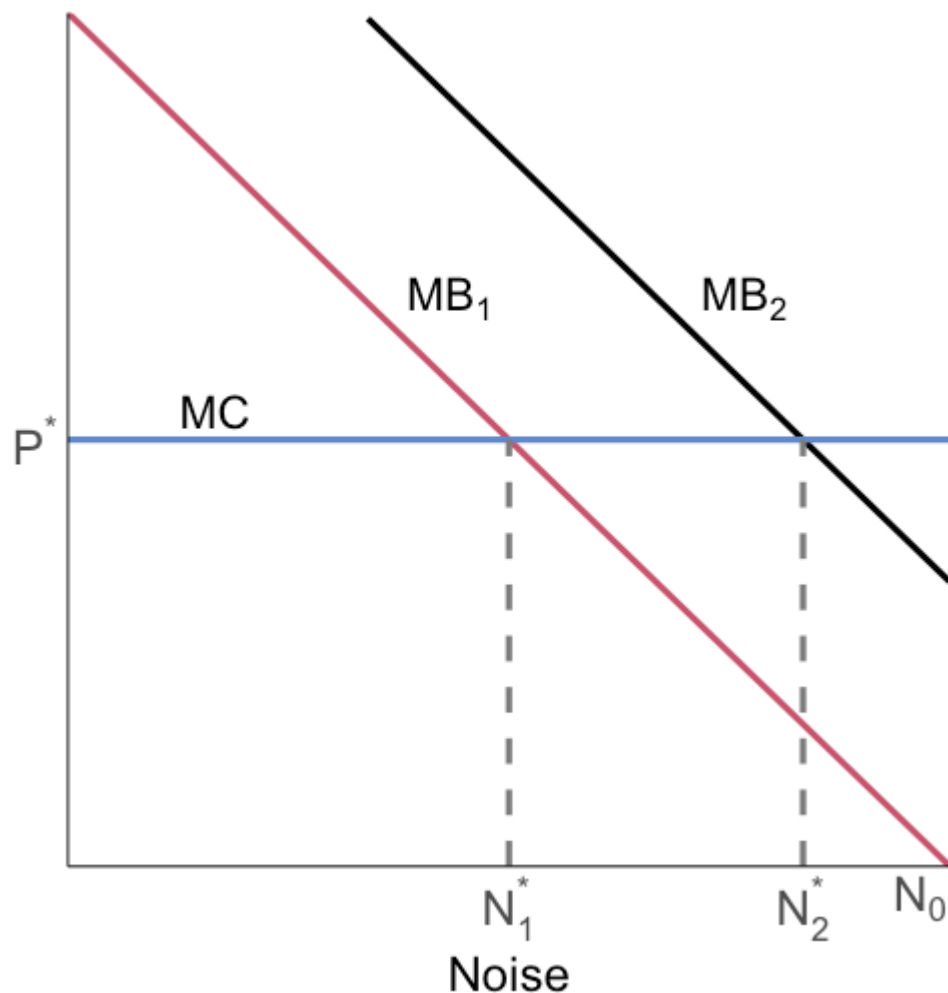
Suppose the confectioner is given the property rights (start at N_0)

Doctor pays confectioner to eliminate noise and move to N_1^*

The confectioner gets some amount of surplus/income

What can the confectioner do with it?

Coase Caveats: Income effects



The confectioner can buy more candy-making machines, increasing its MB from noise

This changes the optimal noise to N_2^*

We have a new equilibrium!

If they contracted to reach N_1^* , it is now **inefficient**

Ways to alleviate transactions cost

One common issue is incomplete information

Disseminating information can make it easier to know each other's costs and benefits which makes beneficial trades more likely to occur

Ways to alleviate transactions cost

Emergency Planning and Community Right-to-know Act (1986) set-up the Toxic Release Inventory (TRI): <http://www.epa.gov/tri/>

Green (“eco”) labeling

- Allows companies to learn which firms took positive steps to reduce pollution, and to reward them in the marketplace
- Firms need to be able to increase demand by enough to offset higher costs and raise profits

Is there a role for the government to be involved in verifying “green” claims?

What really constitutes “organic”?

Coase theorem

While transaction costs can be the downfall of Coasian bargaining, there are costs to government policy:

- Bureaucratic costs
- Government might set the policy incorrectly

Coase and Pigou both have trade-offs to their approaches to solving environmental issues

An example: The Cheshire transaction

The Cheshire transaction

Cheshire, Ohio: a small town with a population 221 before 2002

In the 1970's the town welcomed the construction of a power plant nearby

Gavin Power plant owned by American Electric Power (AEP): 2.6 GW

Enough power for 2 million people (completed 1975)

In the 1990's, property values in the village plummeted

Why would this be the case?

The Cheshire transaction

Acid rain fallout damaged cars, odors nauseated residents and thick plumes of smoke sometimes blocked the sun

In August 2000 the Environmental Protection Agency declared the Gavin plant in violation of the Clean Air Act

A later study found that the air in Cheshire was five times the level necessary to cause an asthma attack

What happened next? Some real world Coasean bargaining

The Cheshire transaction

Fall 2001: the village selected a law group from Washington, DC to pressure the power plant and its owner to heed their concerns and clean up the plant's emissions

Others wanted the plant to compensate them for diminished property value and to address health concerns

April 16, 2002: AEP announced its plan to acquire the incorporated town for \$20 million

September 24, 2002: AEP announces that it has finalized the buyout. About 90 percent of town residents have participated in the buyout offer and have signed the health waivers and the confidentiality agreements

The Cheshire transaction

Property owners in town receive 3.5x assessed value

Outside town: 2x assessed value

Renters receive \$5k for each year lived in Cheshire, up to \$25k

Must sign a health waiver prohibiting them from suing AEP for future health problems

The Cheshire transaction

Must also sign a confidentiality agreement

Cheshire residents over age of 71 able to remain in homes rent free until death

Original population: 221. Current population: <20

Total settlement disbursed by AEP: \$20 million

Attorneys take about 1/3 of settlement money

The Cheshire transaction: was it a good thing?

We know the efficient pollution control decision was made, why?

AEP could have abated instead of compensating! Basically all the involved parties agreed to the contract

Now imagine that AEP Gavin's control costs were low, and the efficient outcome would be to install additional control equipment

Would it matter whether we granted the “right to clean air” to the town or to AEP?

Data centers: Coasean bargaining in the AI boom

Cheshire, twenty years later

The Cheshire transaction was one power plant buying out one small town

The AI boom is running the same experiment hundreds of times at once

Data centers need land, power, and water next to real neighborhoods

Sometimes the operator and the neighbors bargain, sometimes they end up in court

The difference is exactly what our transactions cost model predicts

When bargaining works: Ashburn, Virginia



When bargaining works: Ashburn, Virginia

Developers offered some Ashburn homeowners around \$4 million per house to sell

Why does bargaining break out here?

- **Few parties:** a developer and a handful of homeowners
- **Well-defined property rights:** you own your land, they cannot take it
- **Large surplus:** the land is worth far more under a data center than as housing

Low transactions costs + big gains from trade → deals close: the land moves to its higher-value use and the homeowners are paid to move

When the deal doesn't close: holdouts



When the deal doesn't close: holdouts

The sisters turned down \$26 million for the family farmland

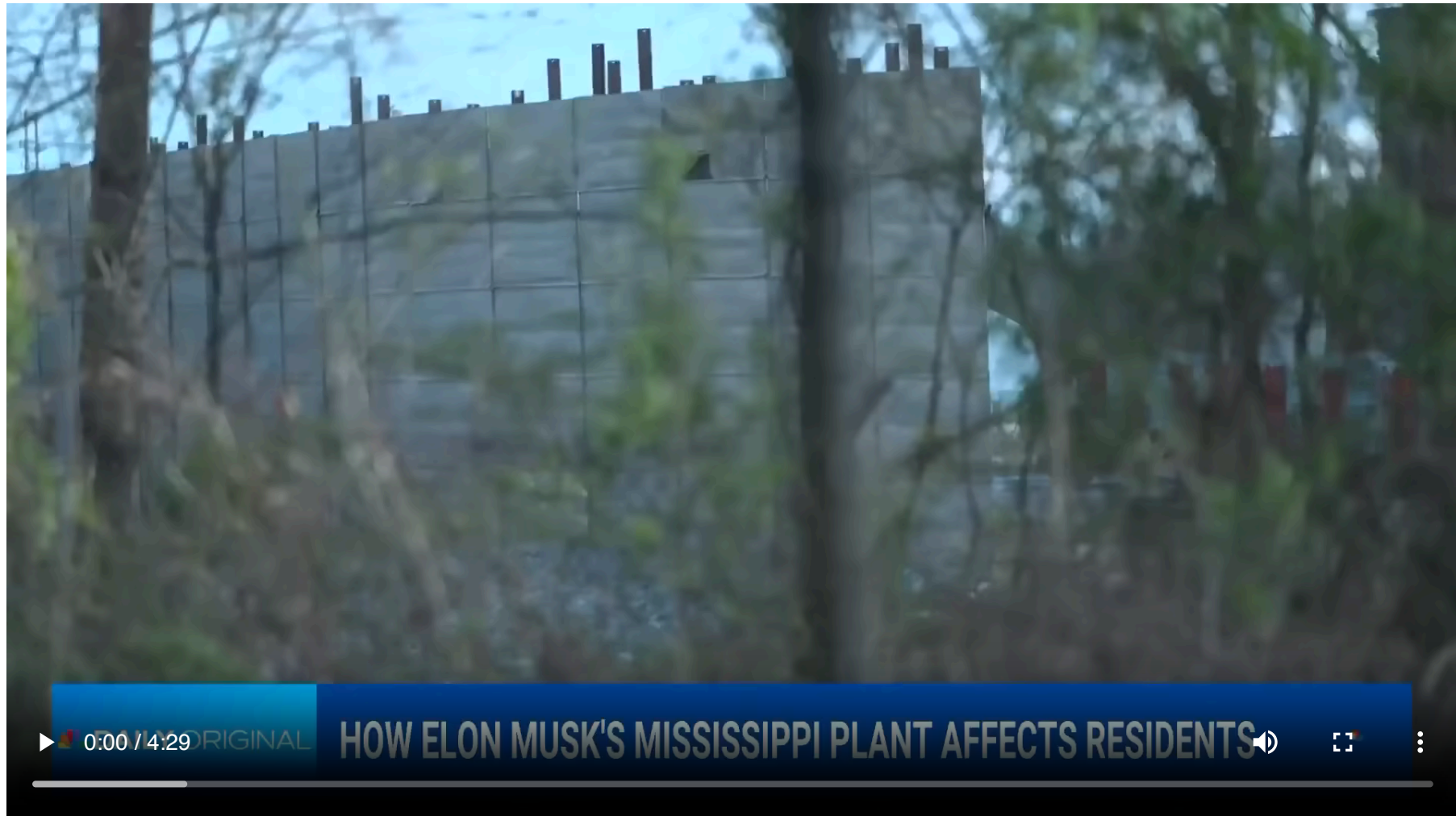
Is this a failure of the Coase theorem?

Not necessarily: if their true willingness to accept exceeds the developer's willingness to pay, **no deal is the efficient outcome**

But land assembly makes bargaining fragile:

- Is a "no" true WTA, or a holdout for a bigger share of the surplus?
- A **contiguous** site gives every owner veto power
- With N owners who must all say yes, transactions costs rise fast

When bargaining fails: Memphis and Southaven



When bargaining fails: Memphis and Southaven

xAI's data center runs dozens of gas turbines next to residential neighborhoods

Why is there no Cheshire-style transaction here?

- **Many parties:** thousands of households breathe the emissions
- **Contested rights:** who owns the air? Permits are disputed, not settled
- **Free riding:** every resident hopes the others fund the fight
- **Information:** health damages are diffuse and hard to document

High transactions costs choke off bargaining → the conflict moves to lawsuits (the NAACP sued under the Clean Air Act) and permit fights

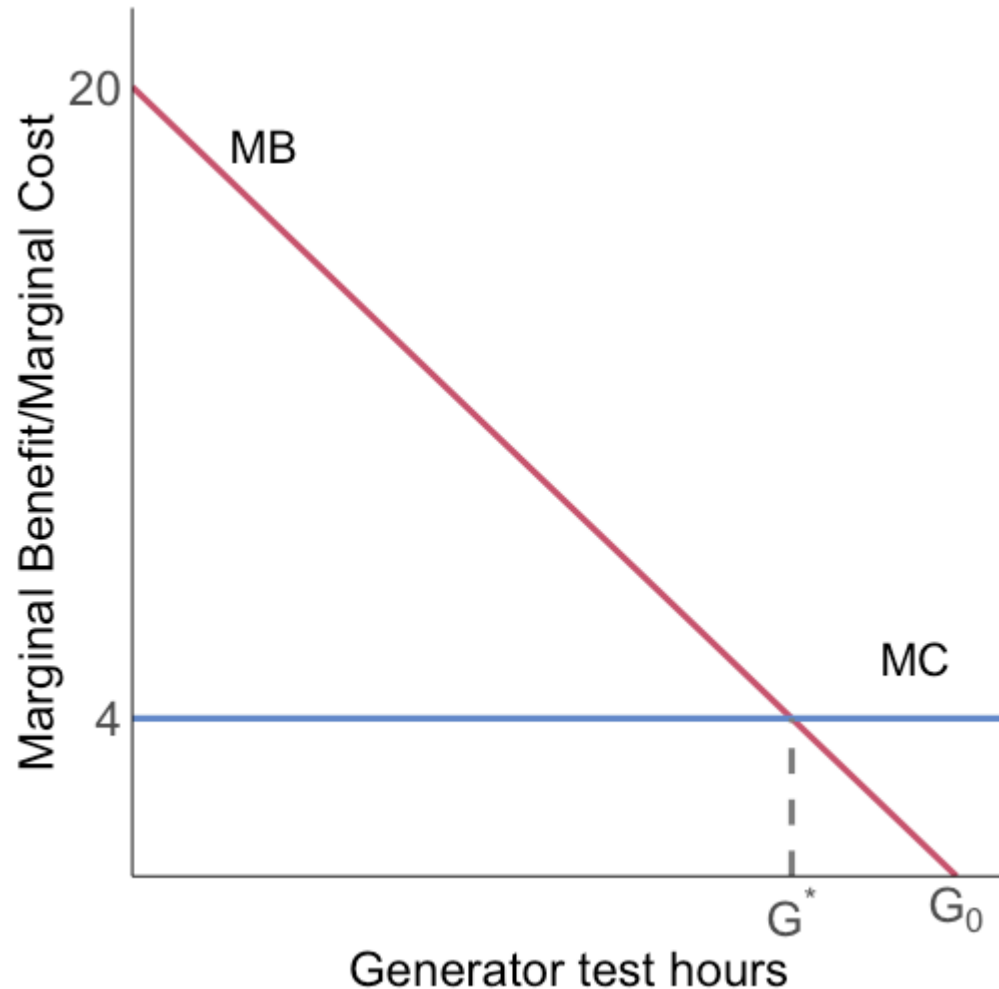
Data centers through Coase's lens

	Ashburn buyouts	Farmland holdout	Memphis turbines
Parties	Few	Few	Thousands
Property rights	Clear (land)	Clear (land)	Contested (air)
Transactions costs	Low	Medium	High
Outcome	Deals close	No deal	Litigation

Same technology, same externalities, three different outcomes

The Coase theorem tells us where to look: **who holds the rights, and how costly is it to strike the deal?**

Your turn: a data center next door



A data center tests its backup generators G hours per month

Marginal benefit to the operator:

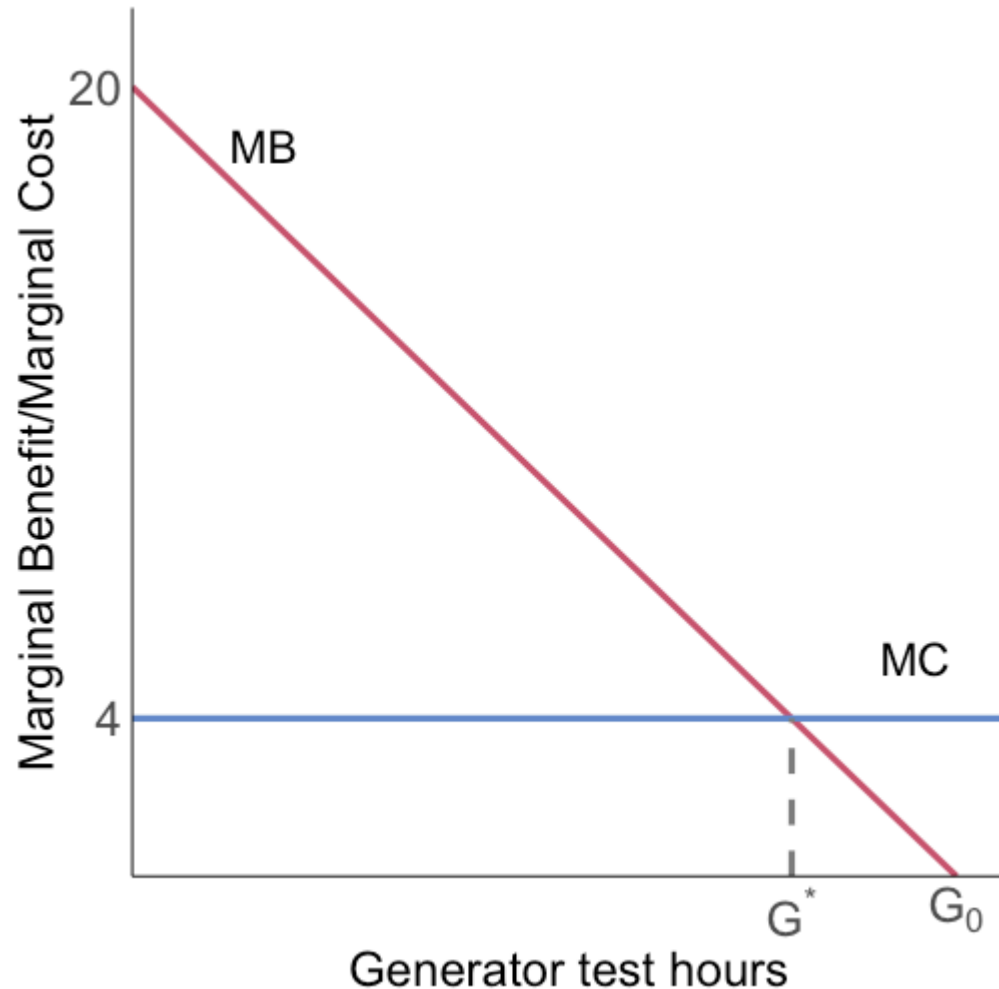
$$MB = 20 - 2G$$

Marginal noise cost to the

residents: $MC = 4$

Left alone, the operator would run $G_0 = 10$ hours; residents want zero

Your turn: a data center next door



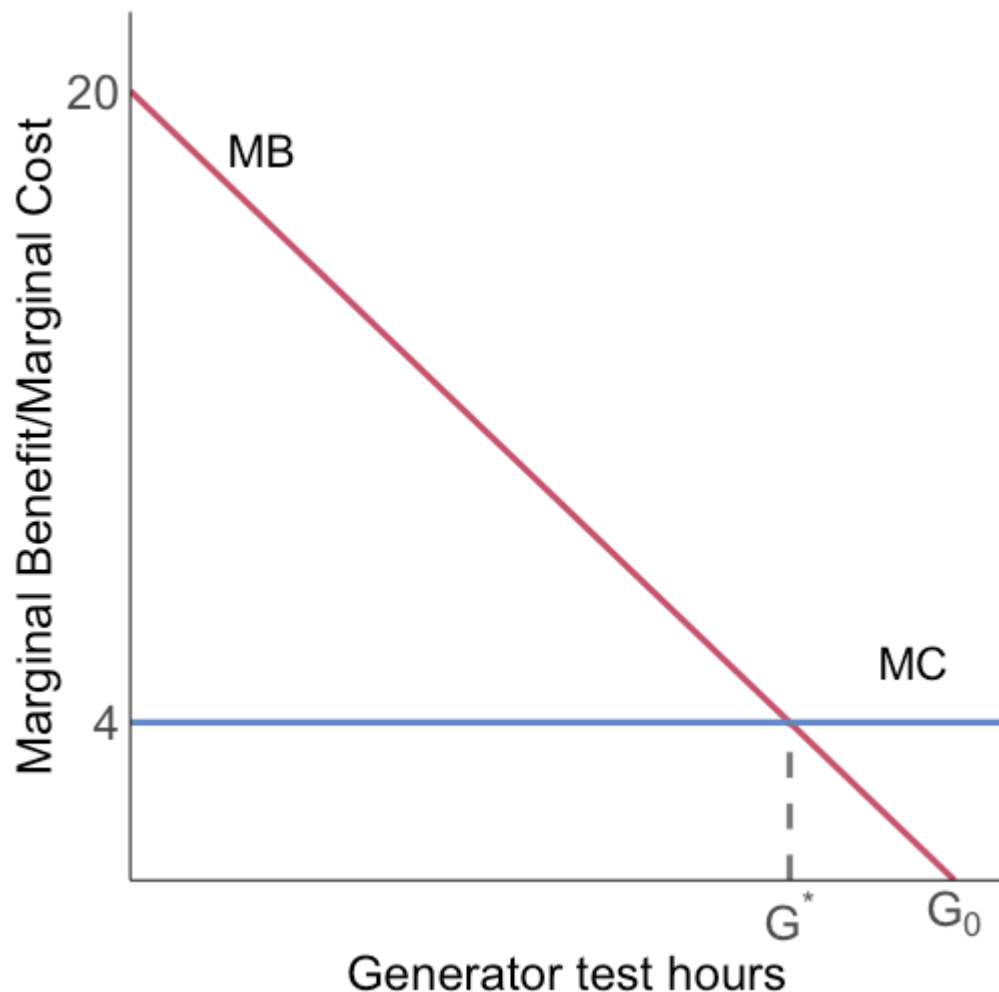
$$MB = 20 - 2G, MC = 4$$

Any deal needs lawyers:
transactions cost $tr = 10$

Your turn:

1. Find the efficient hours G^*
2. Rights to the residents: what payments could work?
3. Under each rights assignment: does a deal happen?

Your turn: the efficient hours

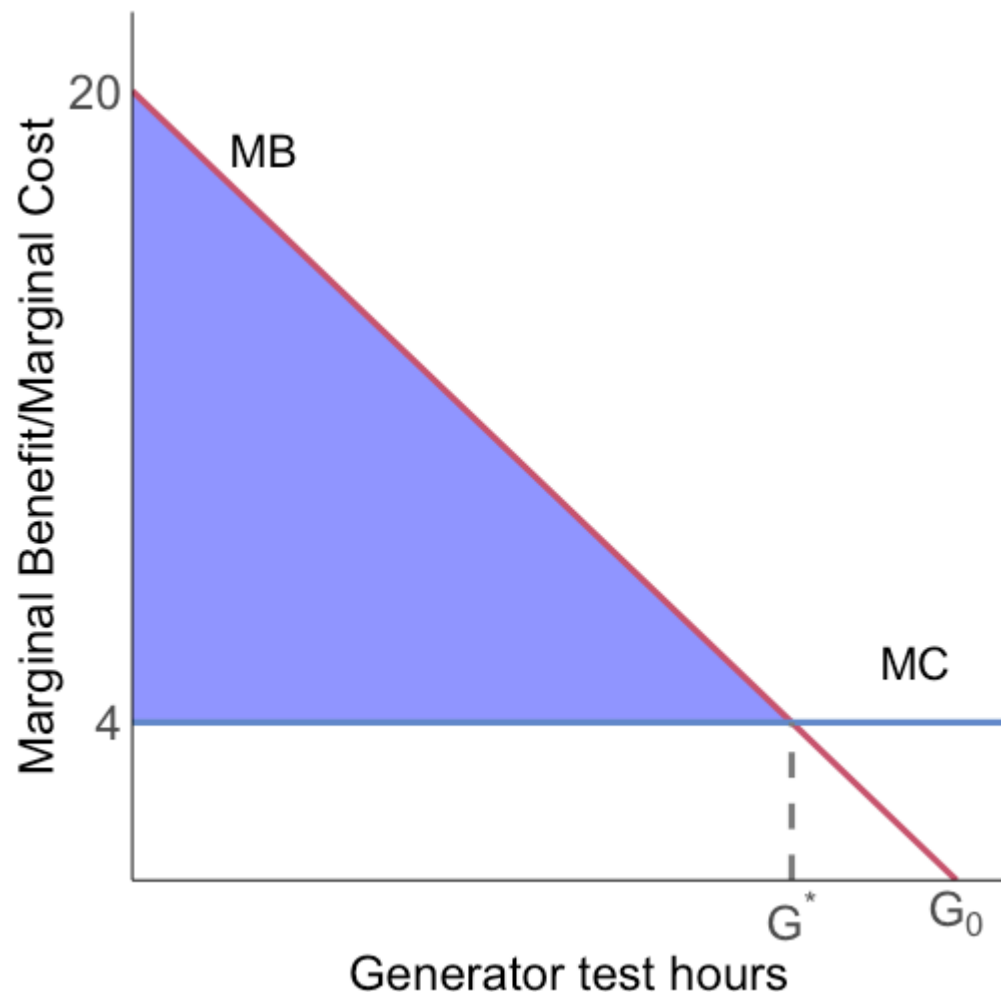


Efficiency: $MB = MC$

$$20 - 2G = 4 \Rightarrow G^* = 8$$

The first 8 hours are worth more to the operator than they cost the residents; the last 2 are not

Your turn: residents hold the rights



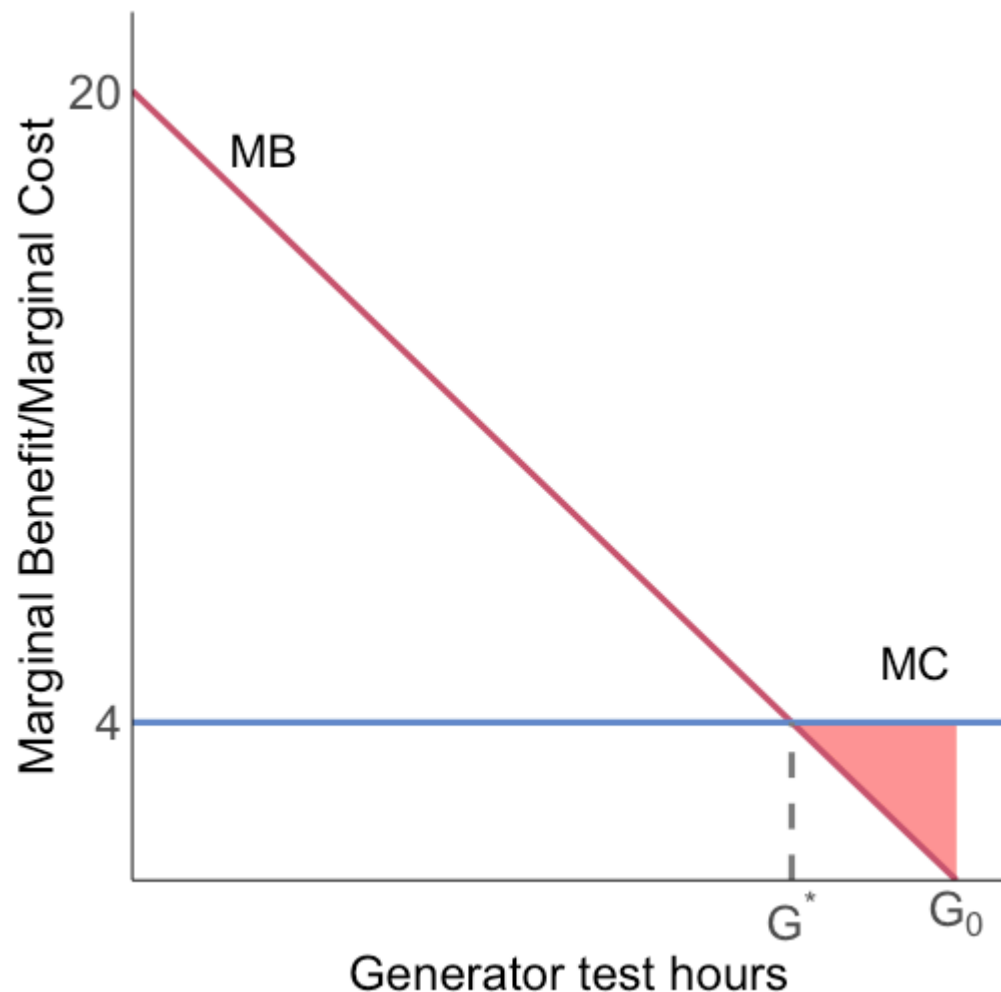
Start at $G = 0$; the operator pays for hours

Residents accept $\theta \geq 4 \times 8 = 32$;
the operator pays at most his benefit, $\theta \leq 96$

Gains from trade: the blue triangle
 $= \frac{1}{2} \times 8 \times 16 = 64$

$64 > tr = 10 \rightarrow$ **the deal happens**
and we reach $G^* = 8$

Your turn: the operator holds the rights



Start at $G_0 = 10$; residents pay to cut hours

Gains from trade: the **red** triangle
 $= \frac{1}{2} \times 2 \times 4 = 4$

$4 < tr = 10 \rightarrow$ **no deal**, and skipping the deal is the right call

With transactions costs, **who holds the rights changes the outcome**

Let's bargain: the data center game

The data center bargaining game

You will be randomly paired: one **data center operator**, one **resident next door**

Bargain in real time over **generator hours** (0 to 6) and a **payment**: send offers, counter, accept, reject, or walk away before the clock runs out

- Round 1: the operator holds the rights (no deal = 6 hours)
- Round 2: the resident holds the rights (no deal = 0 hours)
- Round 3: same as round 2, but any paid deal costs 5 in legal fees

Your payoffs across all three rounds go on the class leaderboard

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The data center bargaining game: what should happen?

If Coase is right, what do we expect to see in the data?

- Rounds 1 and 2 should land on the **same** generator hours
- The **payments** should flow in opposite directions: rights are valuable
- Round 3: think before you deal

We will check the class results against these predictions after we play

Coase in practice

Cap-and-trade is effectively Coase at large scale

It allocates a number of rights to pollute (permits or allowances)

The total number of rights is the cap

Coase tells us that the initial distribution of permits does not matter

The cap-and-trade system will then achieve the efficient outcome